WHAT WORKS IN MARKETING?

An Analysis of Campaign Effectiveness and Customer Response

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15%	0.86	0.72
OVERALL RESPONSE RATE	MODEL AUC SCORE	PRECISION SCORE

01

Executive Summary

This report analyses marketing campaign effectiveness using customer demographics, spending behaviour, engagement patterns, and historical campaign response data. The objective is to identify which customers are most likely to respond to marketing efforts and to provide actionable strategies for improving targeting and return on marketing investment (ROMI).

The analysis reveals that **overall campaign response is low (15%)**, indicating that broad, untargeted marketing approaches are inefficient. Response is highly concentrated within a small segment of customers — particularly those with high spending levels, strong engagement, and prior campaign acceptance.

KEY FINDINGS

1

Spending is the #1 Driver

High-value customers exhibit response rates exceeding 40%, far above the 15% average.

2

History Predicts Future

Repeat responders form a core high-performing segment — prior acceptance strongly predicts future response.

3

Engagement Matters

Recent activity and web interaction are positively associated with campaign acceptance.

4

Income Has Moderate Influence

Top 10% income earners show ~40% response rates versus 11–12% in lower income bands.

STRATEGIC RECOMMENDATION

- Target high-value customers (top spending segments)
- Prioritise previously responsive customers
- Focus on actively engaged, recently active users
- Shift from broad outreach to data-driven targeted strategies

02

Problem Statement & Objectives

Build and evaluate a baseline marketing response model using customer demographics, spending behaviour, channel usage, and historical campaign acceptance to assess whether marketing campaigns are effective, identify which customers are most responsive, and translate insights into actionable recommendations for improving campaign targeting and ROMI.

Research Objectives

Quantify Response Rates: Assess overall marketing response rates and response imbalance (Yes vs No).

Campaign Effectiveness: Evaluate acceptance rates across individual campaigns (Cmp1–Cmp5) and overall responsiveness.

Customer Profiling: Identify how response varies by income, age, household structure, education, and marital status.

Behavioural Drivers: Compare responders vs non-responders on total spend (MntTotal) and channel usage.

Predictive Modelling: Train a baseline response model; evaluate using AUC, precision, recall, and probability thresholds.

Actionable Strategy: Translate analytical results into targeting rules, segmentation strategies, and campaign prioritisation.

PHASE 1 — BASELINE UNDERSTANDING

Overall Marketing Response Rate

The overall marketing response rate is **15%**, with the majority of customers not responding. This indicates a highly imbalanced outcome variable, suggesting that broad, untargeted marketing efforts may be inefficient. Customers with prior campaign acceptance exhibit a substantially higher likelihood of responding, providing early evidence that historical responsiveness is a key driver.

Overall Response Rate Distribution

PHASE 2 — CAMPAIGN-LEVEL EFFECTIVENESS

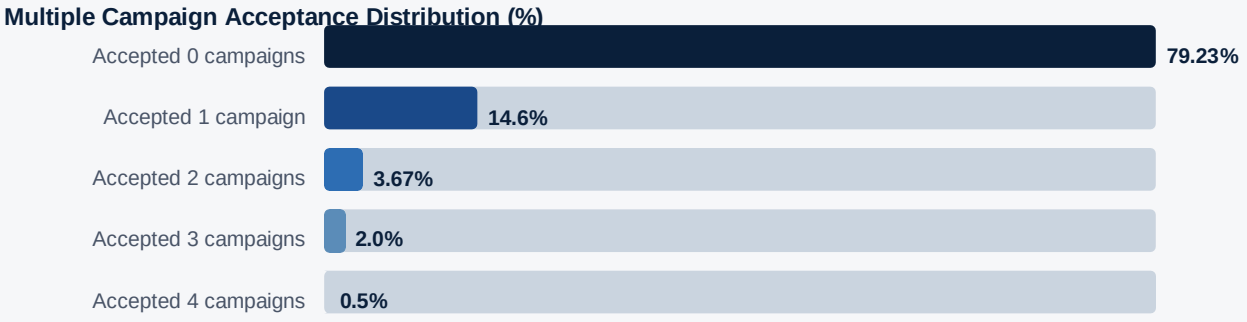
Campaign Performance Analysis

The acceptance rate for each campaign shows clear performance differences. Campaign 4 achieved the highest acceptance (7.44%), while Campaign 2 performed the worst at only 1.36%. Understanding these differences is critical before drawing conclusions about customer responsiveness.

Campaign Acceptance Rates (%)

Distribution of Campaign Responsiveness

Most customers did not respond to any marketing campaign (**79.23% accepted none**). About 14.60% accepted exactly one campaign, while only 6.17% accepted multiple campaigns. This indicates that campaign success is driven by a small group of repeat responders rather than broad customer engagement.


Multiple Campaign Acceptance Distribution (%)
Accepted 0 campaigns
79.23%
Accepted 1 campaign
14.6%
Accepted 2 campaigns
3.67%
Accepted 3 campaigns
2.0%
Accepted 4 campaigns
0.5%

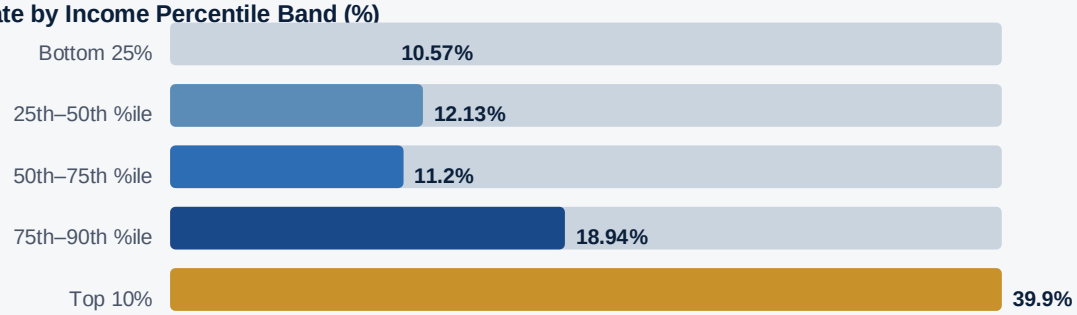
PHASE 3 — CUSTOMER PROFILE DIFFERENCES

Demographic Differences: Responders vs Non-Responders

I. Income and Marketing Response

Response rates remain low and relatively flat across the bottom 75% of income earners (averaging ~11–12%). A clear increase appears in the 75th–90th percentile (~19%). The **strongest response is among the top 10% of income earners, with nearly 40% accepting campaigns**. Marketing effectiveness is therefore driven primarily by higher-income customers.

Response Rate by Income Percentile Band (%)



Statistical significance: chi-square test, $p < 0.001$ (4.70×10^{-24}). Cramer's V = 0.19 — a weak-to-moderate association.

II. Age and Marketing Response

Campaign response rates show only modest variation across age groups. Response declines from the youngest group (17.3%) to middle-aged customers (12.6%), with a slight recovery among the oldest segment (16.4%). Age is **not a strong predictor** of campaign acceptance.

III. Household Composition

Customers without children or teenagers are more likely to accept campaigns. Response rates decline as the number of children increases, with two-child households showing particularly low acceptance. Household composition is a **secondary predictor** compared to income.

IV. Marital Status

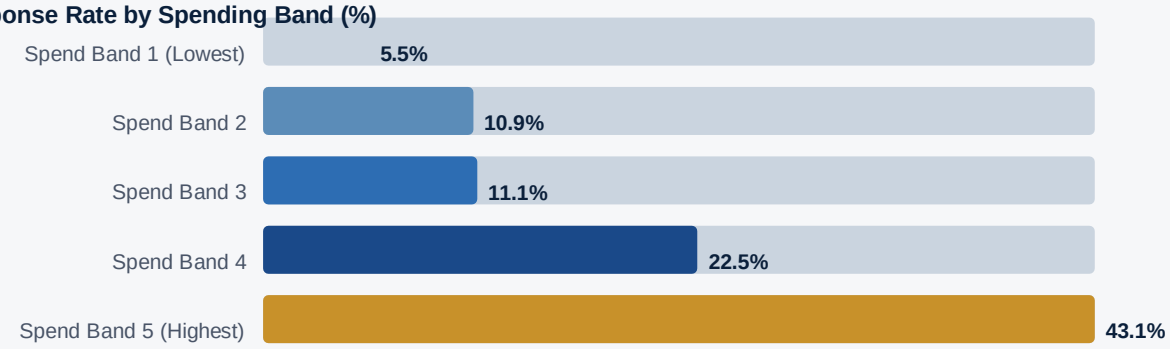
A statistically significant association exists between marital status and response (chi-square = 26.93, $p < 0.001$). Married customers have a higher response rate (33%) vs unmarried (20%). However, Cramer's V = 0.013 — the relationship is **extremely weak in practical terms**.

PHASE 4 — BEHAVIOURAL & VALUE DRIVERS

Spending Behaviour as a Driver of Response

Customer spending shows a **strong, non-linear relationship** with marketing response. Response rates rise from ~5% in the lowest spend group to over 40% among the highest spenders, making spending the single most powerful targeting variable.

Response Rate by Spending Band (%)



Spend Band	Total Customers	Responders	Response Rate
1 (Lowest)	183	10	5.50%
2	828	90	10.90%
3	506	56	11.10%
4	302	68	22.50%
5 (Highest)	202	87	43.10%

IMPLICATION FOR CAMPAIGN DESIGN

- Prioritise Bands 4 and 5 — the top spending segments
- Bands 1 and 2 offer limited return and may not justify broad targeting
- Use total spend as a primary segmentation variable

PHASE 5 & 6 — CHANNEL EFFECTS & EXPERIENCE

Impact of Purchasing Channels on Response

A chi-square test revealed a statistically significant association between web visits and response (chi-square = 21.86, $p < 0.001$). Cramer's V = 0.104 indicates a **weak but meaningful association** — web engagement is positively related to campaign acceptance, though it alone is insufficient for high-precision targeting.

Customer Complaints and Response

No statistically significant association was found between customer complaints and marketing response (chi-square = 0.002, $p = 0.961$). Cramer's V = 0 — prior complaints **do not influence** a customer's likelihood of responding to a campaign.

PHASE 7 — PREDICTIVE MODELLING

Logistic Regression Model Performance

A logistic regression model was trained on an 80/20 stratified split using demographic characteristics, engagement metrics, spending behaviour, and historical campaign acceptance. The model demonstrates strong performance in distinguishing likely responders from non-responders.

87%	0.72	0.29	0.86
ACCURACY	PRECISION	RECALL	AUC

The **high precision (0.72)** indicates that when the model predicts response, it is often correct — critical for efficient budget allocation. The relatively **low recall (0.29)** suggests some potential responders are not identified; however, this is acceptable when the primary goal is precision-driven targeting.

PRACTICAL APPLICATION

- Rank customers by predicted probability of response
- Target top 20–30% of customers only
- Avoid low-probability customers to reduce wasted spend
- Supports data-driven targeting over mass-market campaigns

PHASE 8 — ACTIONABLE INSIGHTS & RECOMMENDATIONS

High-Value Target Customer Segments

Marketing efforts should focus on a narrow, high-potential segment rather than the entire customer base. Four priority groups emerge from the analysis:

1

High Predicted Responders

Rank customers by predicted probability. Target the top 20–30% to maximise ROI and minimise wasted outreach.

2

High-Value Spenders

Spending behaviour is the strongest predictor. Prioritise customers with high MntTotal — revenue-generating, not just volume.

3

Engaged & Active Users

Target customers with recent purchases, higher web visits, and multi-channel activity. Active customers are more receptive.

4

Repeat Responders

Historical acceptors are the strongest predictors of future response. Enroll them in loyalty or VIP programmes.

Strategic Implications for Marketing Optimisation

1. Concentrate budget on high-performing segments.

Marketing performance is driven by a relatively small group of high-value customers. Targeting them intensively generates significantly higher returns than broad-based campaigns.

2. Invest in digital and web channels.

Web visits show a statistically significant (if moderate) relationship with response. Digital campaigns should be personalised and reinforced through frequently used channels. Multi-channel engagement remains key — no single channel should be isolated.

3. Treat repeat responders differently.

Consistent returners drive disproportionate campaign success and require less effort to convert. Offer personalised campaigns, early access, higher-value promotions, and loyalty rewards to this core segment.

SUMMARY & CONCLUSIONS

The analysis demonstrates that marketing effectiveness is driven by a relatively small segment of highly responsive customers. **Broad, untargeted campaigns are inefficient and dilute return on investment.** Instead, organisations should adopt a targeted marketing approach that prioritises high-probability responders, high-value customers, and actively engaged users.

By leveraging predictive modelling and focusing on these segments, firms can significantly improve campaign efficiency, reduce unnecessary marketing costs, and maximise conversion rates. Differentiating strategies for repeat responders, active customers, and lapsed segments will enable more precise and effective marketing execution.

Driver	Association Strength	Recommendation
Total Spend	**Very Strong**	Primary targeting variable
Prior Acceptance	**Very Strong**	Identify & enroll in VIP track
Web Engagement	Moderate	Digital personalisation priority
Income Level	Moderate	Focus on top 10% income band
Age	Weak	Secondary filter only
Marital Status	Very Weak	Not recommended for targeting
Complaints History	None	Ignore in targeting models

Report prepared by Rita Akumu • 21st March 2026 • Marketing Analytics Portfolio